

Intramuscular Botulinum Toxin as an Adjunct to Arthrocentesis with Viscosupplementation in Temporomandibular Disorders: A Proof-of-Concept Case-Control Investigation.

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The reduction in joint load is a potential beneficial factor in managing osteoarthritis of the temporomandibular joint (TMJ). This paper aims to compare the effectiveness of the intramuscular injection of botulinum toxin (BTX-A) as an adjunct to TMJ arthrocentesis plus viscosupplementation with arthrocentesis plus viscosupplementation alone in the management of TMJ osteoarthritis. A pilot clinical retrospective study examined TMJ osteoarthritis treatments. Patients were divided into two groups: Group A received BTX-A injections and arthrocentesis with viscosupplementation, while Group B received only arthrocentesis with viscosupplementation. The study assessed outcomes based on mouth opening (MO), pain at rest (PR), pain at mastication (PF), and masticatory efficiency (ME) at various time points (baseline (T0), 1 week (T1), 2 weeks (T2), 3 weeks (T3), and 4 weeks (T4)) up to 2 months after treatment. The study included two groups, each with five patients. Group A received five weekly sessions of arthrocentesis plus viscosupplementation and a single BTX-A injection during the first arthrocentesis appointment. Group B underwent the five-session protocol of arthrocentesis plus viscosupplementation alone. MO, PF, PR, and ME improved quickly in T2 in both groups, but the improvement was of greater importance over the following weeks and lasted longer in Group A. Arthrocentesis with viscosupplementation associated with BTX-A was found to be more effective than arthrocentesis alone in improving clinical outcomes. This suggests that patients with TMJ osteoarthritis and myofascial pain may benefit from reduced muscle tone and joint load.

Botulinum toxin A attenuates osteoarthritis development via inhibiting chondrocyte ferroptosis through SLC7A11/GPX4 axis.

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Osteoarthritis (OA) is a prevalent joint degenerative disease, resulting in a significant societal burden. However, there is currently a lack of effective treatment option available. Previous studies have suggested that Botulinum toxin A (BONT/A), a macromolecular protein extracted from *Clostridium Botulinum*, may improve the pain and joint function in OA patients, but the mechanism remains elusive. This study was to investigate the impact and potential mechanism of BONT/A on OA in vivo and in vitro experiment. LPS increased the levels of ROS, Fe²⁺ and Fe³⁺, as well as decreased GSH levels, the ratio of GSH / GSSH and mitochondrial membrane potential. It also enhanced the degeneration of extracellular matrix (ECM) and altered the ferroptosis-related protein expression in chondrocytes. BONT/A rescued LPS-induced decrease in collagen type II (Collagen II) expression and increase in matrix metalloproteinase 13 (MMP13), mitigated LPS-induced cytotoxicity in chondrocytes, abolished the accumulation of ROS and iron, upregulated GSH and the ratio of GSH/ GSSH, improved mitochondrial function, and promoted SLC7A11/GPX4 anti-ferroptosis system activation. Additionally, intra-articular injection of BONT/A inhibited the degradation of cartilage in OA model rats. This chondroprotective effect of BONT/A was reversed by erastin (a classical ferroptosis agonist) and enhanced by liproxstatin-1 (a classic ferroptosis inhibitor). Our research confirms that BONT/A alleviates the OA development by inhibiting the ferroptosis of chondrocytes, which revealed to be a potential therapeutic mechanism for BONT/A treating the OA.

Intra-articular Botulinum Toxin Type A: a new approach to treat arthritis joint pain.

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There is a growing need for novel treatments of refractory arthritis joint pain as the aging population is expanding with many patients who are unable to undergo joint replacement surgery. We are studying the efficacy and safety of intra-articular injection of Botulinum Toxin Type A (IA-BoNT/A) into joints with arthritis pain. In several small open label studies, initial effects for IA-BoNT/A were encouraging because two thirds of the patients had more than 50% reduction in joint pain severity that was associated with a significant improvement in function. Importantly no serious adverse effects of IA-BoNT/A were noted. Based on these initial results, we have completed two pilot randomized controlled trials in painful shoulder joints and painful knee joints. In the shoulder study, IA-BoNT/A produced a significant decrease in shoulder pain severity at one month (6.8-4.4 on VAS, $p=.002$) that was also significantly better than the non-significant change after IA-Saline placebo (1.6 unit difference favoring IA-BoNT/A, $p=.014$). In the knee study IA-BoNT/A produced a significant 48% decrease in McGill Total Pain Score at one month ($p=.01$) that was still significant at 3 mo after injection ($p=.002$). There was a strong placebo response in one third of those but the decrease in pain severity was not significant. We are currently conducting a RCT of IA-BoNT/A for painful prosthetic knee joints. Based on these initial studies of IA-BoNT/A we have gone 'back to the bench' to standardize a menu of pain behaviors for mice with acute inflammatory arthritis pain and chronic inflammatory arthritis pain. IA-BoNT/A significantly reduced arthritis joint tenderness (evoked pain score) in acute and chronic inflammatory arthritis and normalized impaired spontaneous wheel running in mice with chronic inflammatory arthritis but not in those with acute inflammatory arthritis. With these models of arthritis and pain behavior methods we will be able to screen potential intra-articular analgesics, define dose response curves and injection schedule, and study the relationships of articular pain and loss of function.

Efficacy of intra-articular botulinum toxin type A in painful knee osteoarthritis: a pilot study.

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To evaluate the efficacy and safety of botulinum toxin type A (BoNT-A) injected intra-articularly in 60 subjects with moderate pain and functional impairment secondary to knee osteoarthritis. The study investigators hypothesized that intra-articular BoNT-A would result in statistically significant improvements in pain and function at 8 weeks. Double-blind, randomized, single tertiary care academic medical center trial with 6-month follow-up. Sixty patients aged 40 years or older with painful osteoarthritis of the knee who had failed physical therapy, medications, and/or injection therapy presenting to the musculoskeletal or orthopedic outpatient clinics at a large tertiary care medical institution. All 60 patients completed 8-week follow-up, but only 32 patients completed the 26-week follow-up. Subjects were randomized to receive a single injection of corticosteroid, low-dose BoNT-A (100 units), or high-dose BoNT-A (200 units). Outcome measures were compared at baseline, 4, 8, 12, and 26 weeks after injection. The primary outcome measure was pain visual analog scale (VAS) at 8 weeks. Secondary outcome measures included Western Ontario McMaster Arthritis Index, Short Form-36 scores, patient global assessment, 40-meter timed walk, and adverse effects. The primary end point was pain VAS score at 8 weeks, which decreased within each group but only reached statistical significance in the low-dose BoNT-A group. In the intra-articular corticosteroid group, VAS decreased from 6.4 $\pm$ 1.8 to 5.4 $\pm$ 2.3 ($P = .15$); for low-dose BoNT-A, from 6.6 $\pm$ 1.9 to 4.5 $\pm$ 2.2 ($P = .01$); and for high-dose BoNT-A, from 6.6 $\pm$ 1.4 to 5.9 $\pm$ 2.4 ($P = .15$). All groups showed statistically significant improvements

in Western Ontario McMaster Arthritis Index scores (pain, stiffness, function) at 8 weeks. No serious adverse events were noted in any group. This pilot study supports a possible role for BoNT-A as a treatment option for symptomatic knee osteoarthritis; however, larger double-blind randomized studies are needed to determine whether BoNT-A is more effective than placebo in this patient population.

Botulinum toxin type A ameliorates adjuvant-arthritis pain by inhibiting microglial activation-mediated neuroinflammation and intracellular molecular signaling.

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Chronic inflammatory pain is a serious clinical problem caused by inflammation of the joints and degenerative diseases and greatly affects patients' quality of life. Persistent pain states are thought to result from the central sensitization of nociceptive pathways in the spinal dorsal horn. Spinal microglia-mediated neuroinflammation plays a pivotal role in the development and maintenance of the central sensitization of chronic inflammatory pain. Botulinum toxin type A (BoNT/A) was recently reported to have analgesic and anti-inflammatory effects. However, the precise mechanism underlying its analgesic effect remains unclear. Although several studies have reported that BoNT/A could regulate neuroinflammation, the reduction of neuroinflammation regulated by BoNT/A in chronic inflammatory pain in experimentally induced arthritis has not been reported. The aim of this study was to investigate whether BoNT/A could alleviate adjuvant-arthritis pain via modulating microglia-mediated neuroinflammation and intracellular molecular pathway. The pain behavioral tests were performed before and after CFA immunization as well as after BoNT/A injection. Western blotting and immunofluorescence staining were used to assess the changes of microglial activation markers (ionized calcium binding adaptor molecule 1, IBA-1) and phosphorylation of P38MAPK (P-p38MAPK) in the lumbar spinal cord. TNF- α and P2X4R gene expression were studied by real-time reverse transcriptase-polymerase chain reaction (RT-PCR). The results showed that (1) the activation of spinal microglia can be continued till 21 days after CFA injection, which suggested its role in the development and maintenance of chronic inflammatory pain. (2) The intra-articular administration of a single effective dose of BoNT/A (5U/10 U) on day 21 after CFA injection significantly reduced nociceptive behaviors and decreased protein overexpression and immunoreactivity for IBA-1 and P-p38MAPK in CFA induced rat. Simultaneously, BoNT/A (5 U) also inhibited the increase in TNF- α mRNA and P2X4R mRNA expression induced by CFA injection. These results suggested that BoNT/A is a potential therapeutic agent for relieving the neuroinflammation that occurs in chronic inflammatory pain by inhibiting the activation of microglial cells and the release of microglia-derived TNF- α . This effect is likely mediated by inhibiting the activation of the P2X4R-P38MAPK signaling pathways in spinal microglial cells.

Application of botulinum toxin in pain management.

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Botulinum toxin has been used for the treatment of many clinical disorders by producing temporary skeletal muscle relaxation. In pain management, botulinum toxin has demonstrated an analgesic effect by reducing muscular hyperactivity, but recent studies suggest this neurotoxin could have direct analgesic mechanisms different from its neuromuscular actions. At the moment, botulinum toxin is widely investigated and used in many painful diseases such as myofascial syndrome, headaches, arthritis, and

neuropathic pain. Further studies are needed to understand the exact analgesic mechanisms, efficacy and complications of botulinum toxin in chronic pain disorders.

Pain behavior measures to quantitate joint pain and response to neurotoxin treatment in murine models of arthritis.

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To evaluate the validity of newly developed pain behavior measures in two murine models of inflammatory arthritis and to determine the ability of these measures to evaluate the analgesic effectiveness of intra-articular (IA) botulinum toxin type A (BoNT/A) for treatment of arthritis pain. Acute inflammatory arthritis was produced in adult female mice by IA injection of carrageenan and chronic inflammatory arthritis by IA injection of CFA. The presence of arthritis was confirmed by the presence of swelling and erythema. A menu of pain behavior measures was devised for quantitating pain in these models including tenderness, and spontaneous nocturnal wheel running. Toxicity due to neurotoxin was measured as gross limb weakness and impaired functional ability during wheel running. Tenderness measures and spontaneous nocturnal wheel-running are valid measures of arthritis pain and are sensitive to the effects of analgesia. Narcotic analgesics are effective, but in fully analgesic doses impair wheel-running. IA BoNT/A is an effective analgesic for chronic arthritis pain, but not for acute arthritis pain. High doses can produce local limb muscle weakness, which impairs wheel-running function. Doses of botulinum toxin that are not toxic retain their analgesic function. Tenderness and spontaneous pain behavior measures are valid and sensitive for the measurement of pain and analgesia in murine models of inflammatory arthritis. Effective narcotic analgesia produces a decline in function in mice similar to that seen in humans. IA neurotoxin is a promising therapy for chronic inflammatory arthritis but may not be effective for acute arthritis pain.

Osteoarthritis of the hip conservative treatment with type A botulinum toxin.

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Osteoarthritis (OA) of the hip is a low-grade inflammatory disease of multiple etiology that has a limited conservative management and insufficiently explored. The application of botulinum toxin type A (BoNT-A) produces in the applied muscle a temporary, delimitable and reversible flaccid paralysis, which applied in a strategic way achieves to reduce mechanical stress and pain. To evaluate the effect on pain perception, functionality and rigidity and changes in flexibility, internal and external rotation, before and 90 days after the application of BoNT-A. Clinical, experimental, longitudinal, open study, with a total of 35 patients and 45 hips treated. BoNT-A of 500 U was applied in iliac, adductor brevis and longus muscles. The Wilcoxon test was used to compare the evaluation scores on days 0-90, observing the decrease in referred pain ($p < 0.0001$), rigidity ($p < 0.002$), improved perceived function ($p < 0.001$) and mobility arcs: flexion, internal and external rotation ($p < 0.0001$). The treatment with BoNT-A provides a conservative and safe option for the management of symptoms and physical restraint caused by hip OA.

Botulinum toxin A dampened inflammatory response in BV-2 microglial cells.

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Our previous studies have demonstrated the analgesic effects of botulinum toxin type A (BoNT/A) in a pre-clinical model of rheumatoid arthritis of the temporomandibular joint, where we proposed that BoNT/A decreases the neurogenic milieu after reaching the subnucleus caudalis. However, it is unknown whether BoNT/A directly regulates microglial cell activity. Therefore, the present study investigates the effects of BoNT/A on a microglial murine cell lineage (BV-2) in different inflammatory conditions. Cellular viability and proliferation were carried out with different concentrations of BoNT/A (ranging from 0.3125 to 20 U/mL) for 24 h. Cells were primed with carrageenan (300 µg/mL) or Lipopolysaccharides (LPS) (20 ng/mL). The gene expression of IL-1 β , IL-6, IL-18, TNF- α , Ikk β , p65, Iba1 were quantified using PCR-RT. The supernatant was used to determine IL-1 β , IL-6, and TNF- α levels. For all data, the significance level was set at 5%. Overall, data analysis revealed that BoNT/A 1.25 U/mL exhibited the greatest effect cell viability and proliferation. In addition, genes associated with inflammatory response in both stimuli (carrageenan and LPS) were downregulated in the presence of BoNT/A. Lastly, BoNT/A mitigates the protein levels of IL-1 β and TNF- α in a time and dose-dependent manner. In conclusion, our results revealed that BoNT/A directly modulates the microglial cells' activities in an inflammatory context, opening new perspectives for using BoNT/A, considering its potential immunomodulatory effect.

Suicide among physicians: Major risk for women physicians.

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The risk of suicide is related to professional activity. Preliminary data suggest that being in the medical profession increases the risk of suicide in women. The objective of this nationwide study is to compare the death rate of physicians due to suicide with that of the general population and to assess the differences based on gender. All physicians and the general population who died by suicide in Spain between 2005 and 2014, both inclusive, were studied. Between these years, the Spanish population grew from 43,662,613 to 46,455,123 persons and from 199,123 to 238,240 number of doctors. The data relating to the deaths of physicians were extracted from the databases of the General Council of Official Medical Associations (CGCOM) and data related to the general population were obtained from the National Institute of Statistics (INE). The variables included in the analyses are gender, age, specialty, place of residence and death, and causes of death according to the ICD-10. The annual mean of physician deaths was 918, with an annual crude rate of 4.8 per 1,000 registered physicians. It is confirmed that physicians have a significantly higher suicide rate (average of 1.3%) than the general population (average of 0.8%) ($p = 0.003$). The comparison of suicide between men and women doctors is significantly higher in women ($X^2 = 53.068$, $p < 0.001$). In addition, if we separate by gender, female physicians have a suicide rate 7.5% higher than women from the general population, although the difference is not significant ($X^2 = 2.614$, $p = 0.107$). . Suicide is higher among physicians than the general population and affects female physicians significantly more.